



4748 - The Ancient Star Formation History of the Isolated, Quenched Ultra-faint Dwarf Galaxy Tucana B

Cycle: 3, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. David J. Sand (PI)	University of Arizona
Dr. Paul Bennet (CoI)	Space Telescope Science Institute
Dr. Denija Crnojevic (CoI)	University of Tampa
Dr. Catherine Fielder (CoI)	University of Arizona
Dr. Michael Gordon Jones (CoI)	University of Arizona
Dr. Ananthan Karunakaran (CoI) (CSA Member)	University of Toronto
Dr. Burcin Mutlu-Pakdil (CoI)	Dartmouth College
Dr. Kristine Spekkens (CoI) (CSA Member)	Queen's University

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Deep TucB observations				
	1	TucB	NIRCam Imaging	(1) TUCANAB

ABSTRACT

We request F090W+F150W NIRCam imaging of Tucana B, a newly discovered, isolated ultra-faint dwarf galaxy beyond the edge of the Local Group ($M_V = -6.9$; $D = 1.4$ Mpc). Tucana B is a unique ultra-faint, as it has likely never had an encounter with the Milky Way's hot halo or tidal field, yet its ground-based observations only display an old (~ 13 Gyr), metal poor ($[Fe/H] = -2$) stellar population, and it lacks any HI gas with which to make new stars -- it is quenched. Reionization is expected to quench many of the smallest galaxies at early times, heating their gas reservoirs and stopping any subsequent star formation. Observations of Tucana B down to the oldest main sequence turnoff will be a definitive test of the role that

reionization plays, identifying whether early star formation stopped around the reionization epoch, or continued up until more recent times, as is seen in more massive field dwarfs. Only JWST can perform these observations well outside the Milky Way's virial radius, where ultra-faint dwarf galaxies are unaffected by ram pressure and tidal forces. Tucana B is currently the only target that can test this prediction of modern galaxy formation theory.

We have designed our observations with potential, future proper motion measurements in mind, which could further elucidate the origins and history of Tucana B with respect to the Local Group.

OBSERVING DESCRIPTION

We will image Tucana B with NIRCcam with the short-wavelength filters F090W and F150W for our primary science goals, while simultaneously obtaining F277W data. Tucana B has a small size on the sky and fits in one NIRCcam detector. We will position it on one of the module B detectors using a postarg-offset. We will use both NIRCcam modules to fully characterize Tucana B and the background.

We will image down to $F090W=29.8$, $F150W=29.0$ with a $S/N=5$, assuming a G5V star spectrum, similar to that expected for main sequence turnoff stars. We have used a MEDIUM8 readout and subpixel dither pattern. We will not cover the detector gaps, and prefer the uniform depth for our science goals. All exposure times have been calculated using the JWST ETC.

We are happy to work with STScI staff if this or any other aspects of our program can be made more efficient.

Proposal 4748 - Targets - The Ancient Star Formation History of the Isolated, Quenched Ultra-faint Dwarf Galaxy Tucana B

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	TUCANAB	RA: 22 47 0.5000 (341.7520833d)		
			Dec: -58 24 27.00 (-58.40750d)		
			Equinox: J2000		
	Comments: Category= <i>Galaxy</i> Description= <i>[Dwarf spheroidal galaxies]</i>				

Proposal 4748 - Observation 1 - The Ancient Star Formation History of the Isolated, Quenched Ultra-faint Dwarf Galaxy Tucana B

Observation	Proposal 4748, Observation 1: TucB										Thu Mar 13 16:00:16 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	TUCANAB	RA: 22 47 0.5000 (341.7520833d) Dec: -58 24 27.00 (-58.40750d) Equinox: J2000 Comments: Category=Galaxy Description=[Dwarf spheroidal galaxies]								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				12	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F090W	F277W	MEDIUM8	8	5	60	12	50763.449	173358	
	2	F150W	F277W	MEDIUM8	5	4	48	12	25124.042	173358	
Special Requirements	Aperture PA Range 80.92542306 to 325.92542306 Degrees (V3 81.0 to 326.0) Aperture PA Range 350.92542306 to 76.92542306 Degrees (V3 351.0 to 77.0) Offset -119.2343702293613 arcsec, -31.868097690449535 arcsec										